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Transcriptome-wide analysis of differential expression in perturbation atlases

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Single-cell CRISPR screens such as Perturb-seq enable transcriptomic profiling of genetic perturbations at scale. However, the data produced by these screens are noisy, and many effects may go undetected. Here we introduce transcriptome-wide analysis of differential expression (TRADE)—a statistical model for the distribution of true differential expression effects that accounts for estimation error appropriately. TRADE estimates the 'transcriptome-wide impact', which quantifies the total effect of a perturbation across the transcriptome. Analyzing several large Perturb-seq datasets, we show that many transcriptional effects remain undetected in standard analyses but emerge in aggregate using TRADE. A typical gene perturbation affects an estimated 45 genes, whereas a typical essential gene affects over 500. We find moderate consistency of perturbation effects across cell types, identify perturbations where transcriptional responses vary qualitatively across dosage levels and clarify the relationship between genetic and transcriptomic correlations across neuropsychiatric disorders.

A foundational approach in modern biology involves measuring the phenotypic effect of genetic variation, occurring naturally or induced experimentally, on gene regulation in cells through differential expression. A particularly scalable approach to differential expression is Perturb-seq, which combines single-cell RNA sequencing (RNA-seq) with a high-throughput CRISPR screen[1–4]. Perturb-seq allows thousands of genetic perturbations to be tested in a single pool of cells, but the number of cells per perturbation (the sample size) is limited, such that many true effects may not be detected.

Human geneticists have contended with a similar issue in genome-wide association studies, where power is also limited. Even when genome-wide significant associations are few, statistical methods can be used to estimate single nucleotide polymorphism (SNP) heritability—a measure of the total genetic effect[5]—and to characterize nearly every aspect of genetic architecture. With data from Perturb-seq, an analogous approach would be to estimate the distribution of differential expression effects, including those that are missed at a chosen significance threshold.

Some existing methods capture aspects of this distribution. The energy distance quantifies the strength of a perturbation after normalization, filtering and projection onto principal components[6,7]. Geneset enrichment analysis ranks perturbations and tests for an enrichment nonparametrically[8]. Rank–rank hypergeometric overlap uses a similar approach to test for a correlation across experiments[9]. iDEA uses a point-normal model for the distribution of differential expression effects to increase association power for the identification

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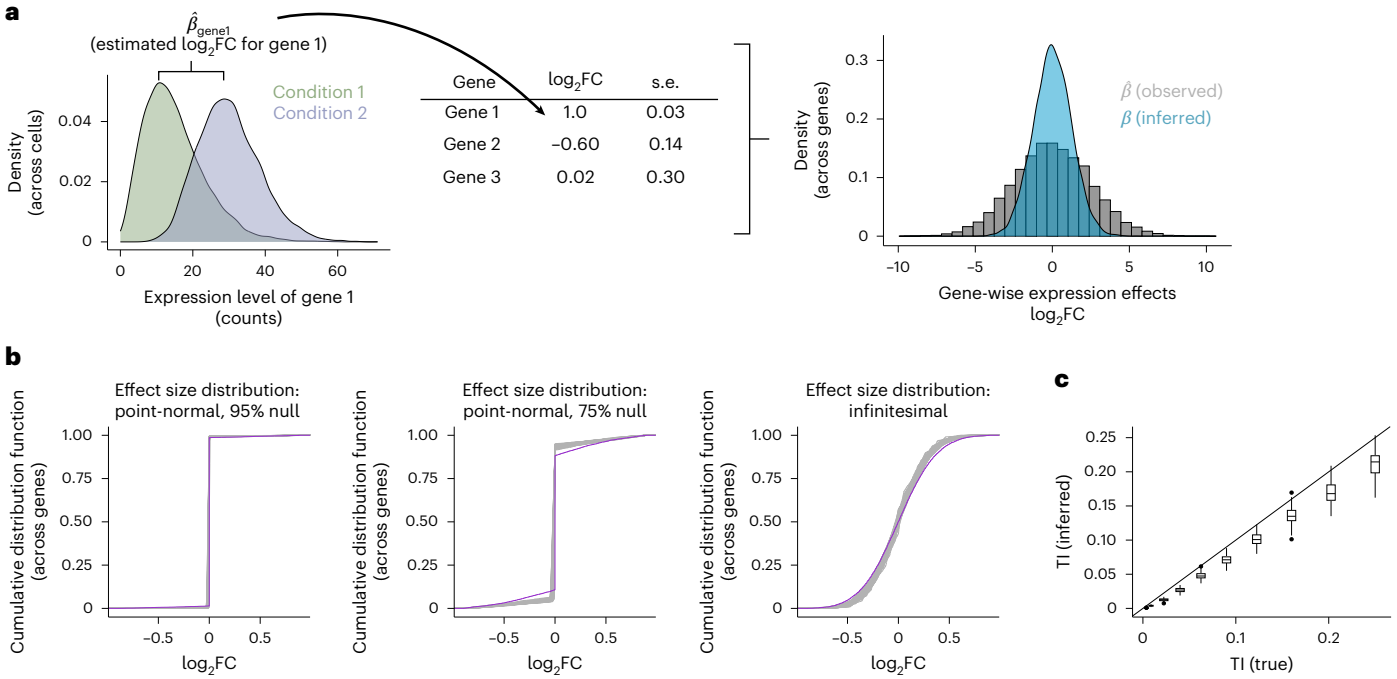


Fig. 1 | Transcriptome-wide analysis of differential expression. a, Schematic for TRADE analysis, starting from condition-wise gene expression counts and ending with estimated distribution of $\log_2$FC. **b**, Estimation of various simulated effect size distributions (point-normal with 95% null, point-normal with 75% null, infinitesimal/normally distributed). Purple traces, true effect size distribution; gray traces, estimated distributions across 100 replicates. **c**, Comparison of estimated and true TI in simulations. Each boxplot is composed of 100 independent simulations. FC, fold change; s.e., standard error.

of individual differentially expressed genes (DEGs)[10]. However, while these approaches are useful, they have conceptual limitations. Some lack a clear estimand (such as the running-sum GSEA statistic); some have ambiguous units (such as the energy distance) and some are biased at finite sample size (such as the proportion of DE genes π in iDEA[11]). Directly estimating the distribution of differential expression effects would address these issues.

Here, we present TRADE—a suite of statistical tools for formally modeling distributions of differential expression effects from RNA-seq experiments, including Perturb-seq. TRADE fits a flexible mixture model to estimated effects and standard errors to estimate the distribution of true differential expression effects. From this estimated distribution, we derive estimators of the total perturbation effect, the effective number of DEGs, geneset enrichments and correlation. We use TRADE to estimate and interpret these features for tens of thousands of genetic perturbations across two new, and three existing massive Perturb-seq datasets[6]. Finally, we use TRADE to compare the effects of perturbations across cell types, to estimate dose–response curves for transcriptome-wide effects, and to estimate the bivariate transcriptomic relationships between neuropsychiatric conditions.

Results

Overview of methods

Consider an RNA-seq experiment comparing two conditions. A conventional differential expression analysis fits a generalized linear model for each gene to estimate the difference in mean expression between conditions, producing a point estimate of the $\log_2$-transformed fold change and a standard error or P value. The point estimate for gene g can be modeled as the sum of a true effect, β_g, and a residual, ϵ_g:

$$\hat{\beta}_g = \beta_g + \epsilon_g$$

Throughout this manuscript, we refer to $\hat{\beta}_g$ as the estimated effect size and β_g as the true effect size. TRADE is a method to estimate the distribution of β from that of $\hat{\beta}$ accounting for sampling variation. This approach distinguishes properties of a cellular response from those of an experiment, such as sample size and sequencing depth, by explicitly accounting for noise in the estimators.

TRADE takes as input differential expression point estimates and standard errors, which we compute using DESeq2 (ref. 12) (Methods) applied to pseudobulk RNA-seq read count matrices (Fig. 1a). TRADE then estimates the distribution of β using ash[13] (Methods). From this distribution, TRADE computes several interpretable features that describe the transcriptome-wide landscape of differential effects. One key estimand is $\mathrm{Var}(\beta)$—the variance of the effect size distribution—which we call the transcriptome-wide impact (TI) (Methods). TRADE also estimates geneset enrichments, correlations between different experiments and the effective number of DEGs, all of which are unbiased at finite sample size (Supplementary Figs. 1, 12, 14 and 16 and Methods).

We tested our approach in simulations (Fig. 1b and Supplementary Note 1). The inferred distributions closely matched the simulated true distributions, except that they slightly overestimated the density at zero. Estimates of the TI (Fig. 1c) were robust, with slight downward bias. The downward bias results from inadequate power to detect very small effects at finite sample size, even aggregating power across genes; at increased sample size, the bias disappears (Supplementary Fig. 1). We explored TRADE estimates generated from limmaTrend[14] finding that this resulted in more severe downward bias (Supplementary Note 2).

TI of 9,866 genetic perturbations

Replogle et al.[6] performed genome-scale Perturb-seq with CRISPR interference (CRISPRi), which inhibits target gene transcription by recruitment of a dCas9-linked repressive KRAB domain, generating three datasets: K562-GenomeWide (perturbations of all 9,866 expressed genes in the K562 chronic myelogenous leukemia cell line), K562-Essential (2,057 common essential gene perturbations in the same cell line) and RPE1-Essential (2,393 common essential gene

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Table 1 | Characteristics of Perturb-seq datasets analyzed

Dataset (technology)	Reference	Cell type	Perturbed geneset	Number of perturbations	Median no. of cells per perturbation
K562-GenomeWide (CRISPRi)	Replogle et al.[6]	K562	All expressed genes	9,866 genes	178
K562-Essential (CRISPRi)	Replogle et al.[6]	K562	Essential genes	2,057	121
RPE1-Essential (CRISPRi)	Replogle et al.[6]	RPE1	Essential genes	2,393	72
Jurkat-Essential (CRISPRi)	New	Jurkat	Essential genes	2,393	83
HepG2-Essential (CRISPRi)	New	HepG2	Essential genes	2,393	45
K562-Titration (CRISPRi)	Jost et al.[25]	K562	Essential genes	25 genes, 128 guides	143
SOX9-Titration (dTAG)	Naqvi et al.[26]	CNCC	SOX9	Five dTAG degron concentrations	Seven bulk samples per concentration
Polycomb-Titration (dTAG)	Weber et al.[27]	mES cell	RING1b, EED (simultaneous)	Four dTAG degron concentrations	Four bulk samples per concentration

CNCC, cranial neural crest cells; mES cell, mouse embryonic stem cells.

perturbations in a retinal pigmented epithelium cell line). We performed two additional large-scale Perturb-seq experiments targeting common essential genes in Jurkat and HepG2 cell lines: Jurkat-Essential (2,393 essential gene perturbations in a T cell leukemia cell line) and HepG2-Essential (2,393 essential gene perturbations in a hepatocellular carcinoma cell line) (Methods). Key features of these datasets are summarized in Table 1. In each dataset, we estimated pseudobulk differential expression effects for each perturbation with DESeq2, summing cell-wise expression counts in each batch for perturbed and unperturbed cells (that is, the number of observational units is twice the number of batches) (Methods).

The number of cells per perturbation varies widely due to technical features of pooled screening. For example, in the K562-Essential dataset, 108 cells received *GATA1* knockdown versus just seven for *EIF4A3*. Without accounting for this, both perturbations apparently produce similar effect size distributions. However, TRADE accounts for the difference in sample size and estimates that, whereas *GATA1* knockdown has a large transcriptional effect (TI = 0.4; $P = 9 \times 10^{-12}$), *EIF4A4* knockdown lacks evidence of broad transcriptome-wide effects (TI = 0.004; $P = 0.12$) (Fig. 2a,b). More generally, we observed larger discrepancies between observed and estimated effect size distributions for perturbations with lower sample size (Supplementary Fig. 6).

We estimated the fraction of TI that was explained by false discovery rate (FDR)-significant effects (Supplementary Tables 2–6). In the K562-GenomeWide experiment, 36% of TI was explained by FDR-significant effects (Fig. 2c); in K562-Essential, 18%; in RPE1-Essential, 35%; in Jurkat-Essential, 13% and in HepG2-Essential, 14%. These estimates highlight the value of our threshold-free approach.

We performed several secondary analyses. We confirmed that our TI estimates were minimal in a negative control analysis of non-targeting guide RNAs (Supplementary Fig. 7). We conducted a down-sampling analysis, repeating our analysis of the K562-GenomeWide experiment using only 50% of the 10x Genomics GEM groups. Whereas the signal in significant genes decreased substantially, our estimate of the TI decreased only slightly (Fig. 2c), consistent with the slight downward biased observed in simulations (Supplementary Fig. 1). For well-powered perturbations with estimated TI greater than around 0.01, there was almost no decrease (Supplementary Fig. 8). We compared TI estimates with sample size in the K562-Essential experiment, and found that, in general, we were able to identify perturbations with significant TI across a wide range of sample sizes (Supplementary Fig. 9). Similarly, examining perturbations that are shared between the K562-GenomeWide and K562-Essential experiments, we found that estimates of TI were far more consistent across experiments than the number of significant DEGs (TI $R^2 = 59.7\%$; number of DEGs $R^2 = 28.4\%$; Supplementary Fig. 10).

Our analyses suggest that significant genes do not capture the bulk of TI. How many genes are required to do so? We defined the effective number of DEGs (π_{DEG}) as a function of the kurtosis of the effect size distribution, following the approach described previously[15]. This quantity captures the evenness of differential expression across the transcriptome, without making an arbitrary distinction between zero and nearly zero effects (Supplementary Fig. 11). We validated our estimation procedure for π_{DEG} in simulations (Supplementary Fig. 12) and in a downsampling analysis (Supplementary Fig. 13).

For the K562-GenomeWide experiment, the median π_{DEG} was 45, suggesting that typically, tens of genes are required to explain the bulk of the TI (Fig. 2d). Some genetic perturbations had much larger π_{DEG}; in particular, knockdown of essential gene perturbations in all four cell types analyzed had median π_{DEG} greater than 500 (Fig. 2d). In a simplified model where effects are either null or normally distributed with some variance σ^2, π_{DEG} equals the number of nonnull effects, σ^2 is the ratio between the scaled TI and π_{DEG}, and σ can be interpreted as the average nonnull $\log_2$FC (Supplementary Note 3). We find that σ is stable across genes and cell types, with values mostly in the range [0.1,1] (Supplementary Note 3).

Two types of gene set enrichment

Some sets of genes may produce greater-than-average TI when perturbed, and others may be enriched for differential expression response to perturbations of other genes. We stratified genetic perturbations by features of the targeted genes including: level of expression, effect on cellular growth (essentiality)[16], level of selective constraint in gnomAD[17] and subcellular localization of their protein product[18] (Methods). We quantified two types of enrichment: (1) the perturbation impact enrichment, which quantifies enrichment for effects on other genes and (2) the perturbation response enrichment, which quantifies enrichment for differential expression in response to perturbations (Supplementary Table 7 and Methods). Means of enrichments across perturbations are shown in Fig. 3. In simulations, we confirmed that the TRADE perturbation impact and response enrichment estimators are approximately unbiased, including for small genesets (Supplementary Fig. 14), and we performed further validation with positive-control genesets[19,20] (Supplementary Fig. 15). Genes with more efficient CRISPRi knockdown were not enriched for perturbation impact, suggesting that enrichments are not driven by differences between guides (Supplementary Fig. 15).

We found that constrained genes, which are depleted of loss-of-function variation in the general population, are enriched for perturbation impact by 1.6-fold, consistent with their functional importance (Fig. 3). On the other hand, they are depleted strongly for perturbation response, by 0.4-fold, suggesting that, across genes, population-level

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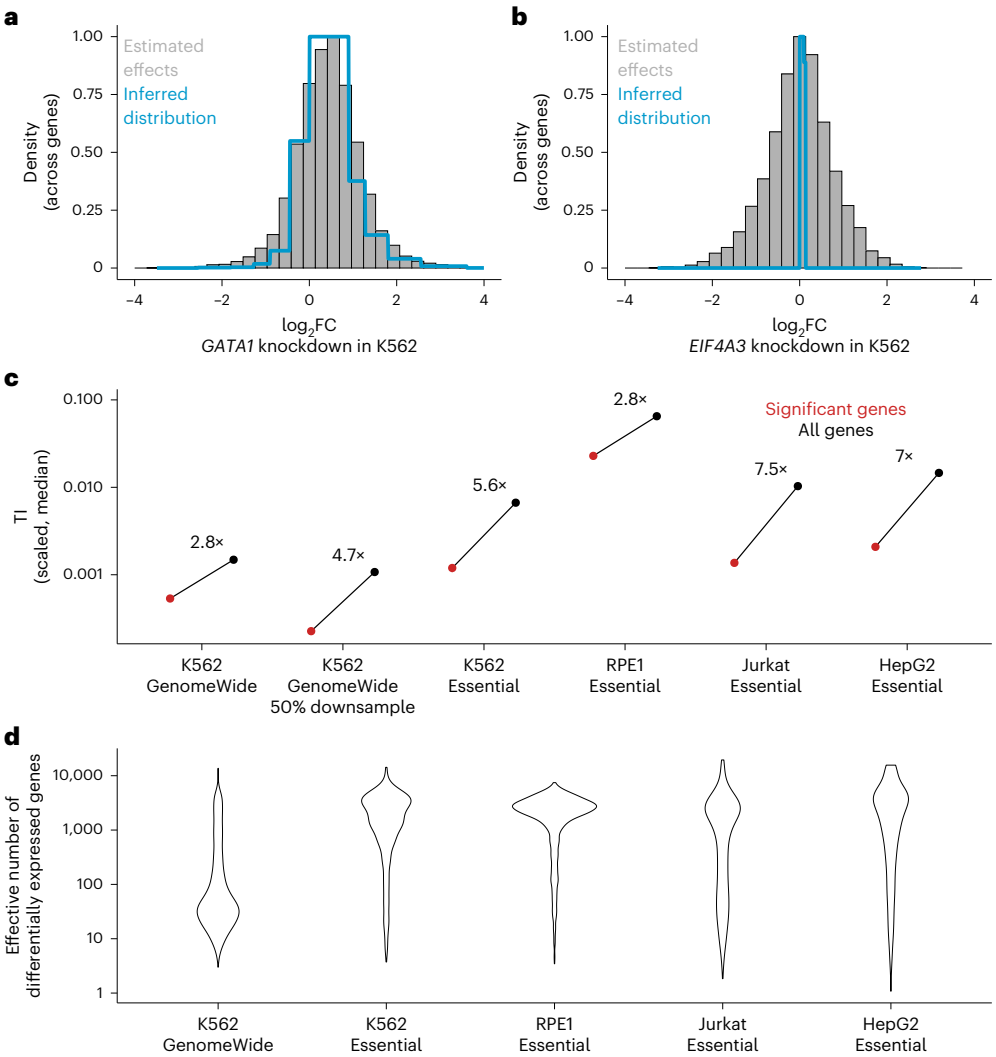


Fig. 2 | Transcriptome-wide analysis of genome-wide Perturb-seq. a, Examples of empirical $\log_2$FC distribution and TRADE inferred distribution for perturbation of *GATA1* in cell line K562. **b**, Similar for perturbation of *EIF4A3* in K562. **c**, Comparison of TI in significant and all genes in Perturb-seq experiments; *y* axis values correspond to TI estimates scaled by the number of measured genes. **d**, Effective number of DEGs (π_{DEG}) across Perturb-seq datasets, for perturbations with nominally significant TI (Methods).

constraint is mirrored by regulatory robustness. Similarly, genes with a strong growth effect in K562 cells (roughly, those that are essential in culture) are enriched strongly for perturbation impact, by 4.2-fold, while being depleted for perturbation response by 0.7-fold. By contrast, genes that are highly expressed in K562 cells are enriched strongly for both perturbation impact (2.3-fold) and perturbation response (4.4-fold), supporting a correlation between absolute expression and functional importance. We observed only a modest perturbation impact enrichments for genes that were localized to the nucleus (1.3-fold), despite their direct role in transcriptional regulation; cytoskeleton-localizing genes were modestly depleted of perturbation impact (0.7-fold).

Consistency of transcriptome-wide effects across cell types

The effect of perturbing a gene may vary across cell types, particularly if it participates in cell-type-dependent functions. These perturbation effects may vary both in magnitude and in which genes are affected. Using data from common essential gene perturbations in the four cell lines (Table 1), we (1) compared TI across cell lines and (2) estimated the correlation between differential expression effects from each experiment using a bivariate extension of TRADE. We refer to this quantity as the 'TI correlation' (Methods) and demonstrate that correlation inference with TRADE is unbiased (Supplementary Fig. 16).

As expected, TI was correlated across cell types (average correlation, 0.62; Supplementary Fig. 17). On average, TI was larger in the RPE1 cell line than in the other three cell lines. A few perturbations did have greater-than-expected effects in specific cell types (Fig. 4a). Using a liberal threshold (Methods), we identified 241 such perturbations (K562, 47; RPE1, 118; Jurkat, 10; HepG2, 66) (Supplementary Table 8). Some of these perturbations are known to be indispensable for their corresponding cell type, including *GATA1* for erythroid cells such as K562 (ref. 21) and *HMGCR* for T cells such as Jurkat[22], but most had no previously documented explanation for their cell-type-dependent effects. As this dataset targeted primarily common essential genes which are expected to be important for growth across most cell types, there are expected to be many more examples of cell-type-specific effects in a larger cellular perturbation atlas.

Before computing correlations between different cell types, we first compared differential expression effects of the same genetic perturbations in repeated experiments (K562-GenomeWide and K562-Essential) (Supplementary Table 9). The median correlation between logFC point estimates—not using TRADE—was only 0.16,

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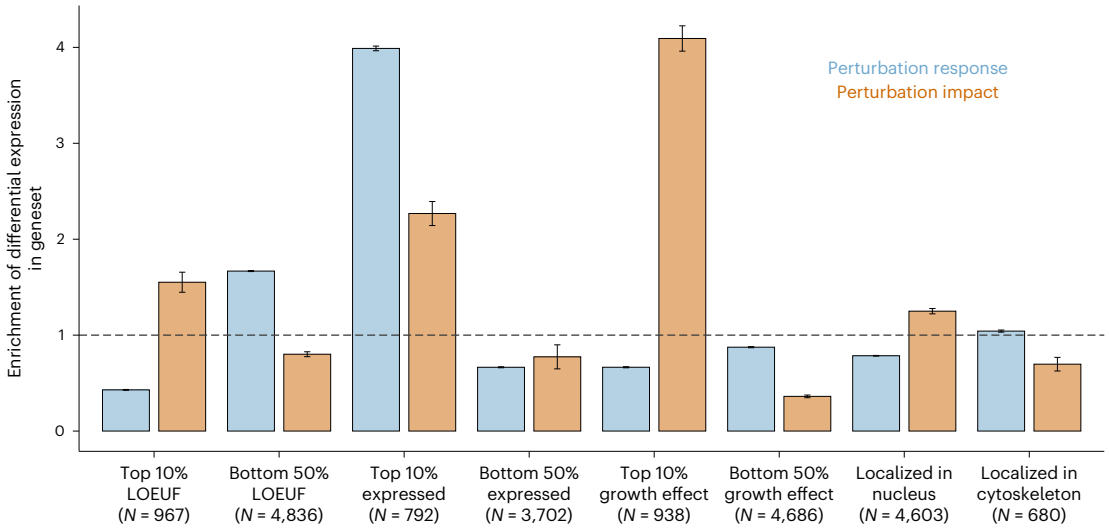


Fig. 3 | TRADE-derived enrichment estimates. Blue bars indicate the perturbation response enrichment, or the enrichment of differential expression in response to perturbations. Tan bars indicate perturbation impact enrichment, or the enrichment of effects on other genes when genes in that geneset are perturbed. The size of each geneset is shown below the *x* axis labels. Error bars denote s.e.m.

apparently suggesting low replicability (Fig. 4b). However, the median TI correlation between replicates was 0.90, implying excellent replicability (Fig. 4b). This difference underscores the value of modeling sampling variance when estimating effect size correlations (as uncorrelated sampling variation causes downward bias in correlation estimates; Supplementary Note 4). A few perturbations did have low between-experiment TI correlations; most of these had very low TI and, thus, their correlations are expected to be noisy (Supplementary Fig. 18).

We used TRADE to estimate the inter-cell-type TI correlation, or the correlation of transcriptome-wide effects between cell types, for perturbations of 2,053 shared essential genes across K562, RPE1, Jurkat and HepG2 (Supplementary Table 10). Because these correlations are not defined in the setting of null TI, we restricted our analysis to 1,660 perturbations with significant TI in all four cell types, using a very liberal threshold ($Z > 0.5$, roughly $P < 0.3$). The TI correlation was highest between K562 and Jurkat (median, 0.74) and between HepG2 and RPE1 (median, 0.75) (Fig. 4c and Supplementary Fig. 19). K562 and Jurkat are hematopoietic cell lines that are p53 mutant and grow in suspension, while HepG2 and RPE1 are epithelial cell lines that are p53 wild type and are adherent. There were weaker correlations between K562 and HepG2 (median, 0.64) and between Jurkat and HepG2 (median, 0.69), and substantially weaker correlations between K562 and RPE1 (0.40) and Jurkat and RPE1 (0.56). A potential limitation of this analysis is its reliance on perturbations that are essential in all four cell lines; however, inferred correlations did not vary strongly with essentiality (Supplementary Fig. 20).

Across perturbations, we observed two patterns of inter-cell-type TI correlations (Fig. 4d). Some perturbations, such as knockdown of *DDX41*, had high correlations across all four cell types (Fig. 4e). Other perturbations, such as knockdown of *NIFK*, had much higher correlations within the pairs K562/Jurkat and RPE1/HepG2 than other cell-type pairs (Fig. 4e). Clustering these perturbations with a Gaussian mixture model (Methods), we found that 44% of perturbations belonged to a cluster with selectively higher correlations between more similar cell types (cluster 1, mean correlation within similar cell-type pairs: 0.61; outside similar pairs: 0.35), whereas 56% of the perturbations belonged to a cluster with high correlations across all cell types (cluster 2, mean correlation within similar cell-type pairs: 0.75; outside similar pairs: 0.66). We observed that genes in cluster 1 (that is, with cell-type-preferential correlations) had differences in TI across cell types that were approximately twice as large as genes in cluster 2 ($P = 4 \times 10^{-8}$; paired *t*-test). This indicates that when a gene has a cell-type-specific function, perturbing that gene in the relevant cell type causes not only a larger magnitude of effect but also a different pattern of effects across genes (Supplementary Fig. 21).

Dosage sensitivity of TI

In the experiments analyzed above, CRISPRi guide RNAs were engineered to maximize ontarget knockdown, but the effects of a perturbation may vary with dosage, and dose–response curves can yield insight into gene regulation and guide therapeutic design[23,24]. We applied TRADE to data from three experiments. Jost et al.[25] titrated gene expression with CRISPRi with attenuated guide RNAs. Naqvi et al.[26] and Weber et al.[27] directly depleted protein levels with the dTag degron system, which can be tuned by titrating a small molecule[28]; this approach has lower throughput but controls protein (as opposed to transcription) levels directly. We quantified (1) the magnitude of the TI as a function of dosage and (2) the correlation of these effects between each pair of dosages.

As expected, stronger perturbations had consistently larger TI (Fig. 5a). For dTAG depletion of SOX9 (Supplementary Table 11) and Polycomb (Supplementary Table 13), the TI dosage-response curve was nonlinear. Weak-to-moderate perturbations of these proteins caused relatively small transcriptome-wide effects, whereas strong perturbations caused disproportionately large transcriptome-wide effects. These genes are haploinsufficient (pLI = 1 (ref. 17)) indicating that 50% depletion is deleterious; our results suggest that stronger depletions nonetheless produce progressively larger cellular effects. We generally observed similar dose–response curves for CRISPRi knockdown of 25 essential genes in K562 cells, with varying degrees of nonlinearity (Fig. 5a, Supplementary Table 15 and Supplementary Fig. 22).

We next quantified the TI correlation between dosage levels for each perturbation (Fig. 5b). For dTag depletion of SOX9 (Supplementary Table 12) and Polycomb (Supplementary Table 14), TI correlations decayed smoothly with the difference in dosage, and the smallest and largest perturbations were only moderately correlated ($r = 0.60$, 0.48), implying that weak and strong perturbations have qualitatively different transcriptional consequences. The CRISPRi knockdown of essential genes produced a range of patterns (Supplementary Table 16 and Supplementary Fig. 23). For example, the response to *BCR* knockdown was highly correlated across all dosage levels, despite substantial

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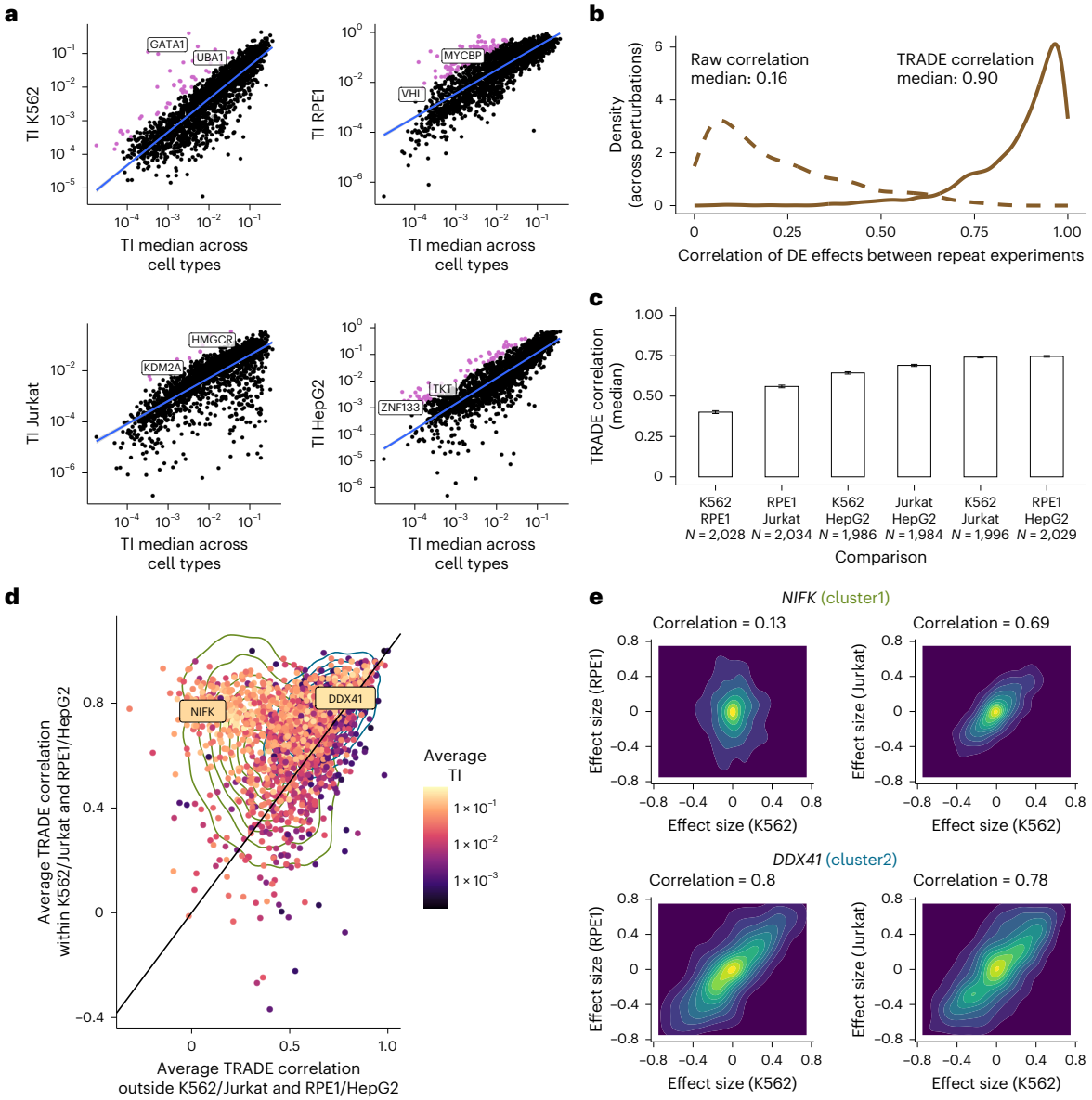


Fig. 4 | Correlation of differential expression across cell types. a, TI of gene perturbations in each cell type versus the median across cell types, with outliers (pink) defined as being more than 1.64 s.d. away from the fit line. **b**, Correlation of differential expression (DE) effects across replicate perturbations in K562. Dotted line, raw correlation; solid line, correlation estimated with TRADE. **c**, Median correlation of perturbation effects for common essential genes for each pair of cell types. The number of perturbations included in each estimate are shown below the *x* axis labels. Error bars denote standard error of the median. **d**, Comparison of effect size correlation strength within similar cell types and outside of similar cell-type pairs. Densities from Gaussian mixture models are overlaid, showing cluster 1 (green) and cluster 2 (blue) **e**, Examples of inferred joint effect size distributions across key cell-type comparisons for perturbations of *DDX41* and *NIFK*.

differences in the magnitude of responses (Fig. 5b). The response to *GATA1* knockdown was highly correlated among all but the weakest perturbation, which was only moderately correlated with the strongest perturbation (Fig. 5b).

We interpret these correlations as a readout of how the dose–response curve varies across target genes. If all downstream genes have identical response curves (up to multiplication by a constant), then the effect of a partial depletion is a fixed fraction of the effect of a full depletion, leading to a cross-dosage correlation of exactly one (Fig. 5c and Supplementary Note 5). However, if the response curve varies between target genes, the correlation is less than 1, to an extent that depends on the variability of response curves (Fig. 5c and Supplementary Note 5). Indeed, Naqvi et al.[26] found that a subset of SOX9 targets are sensitive at partial dosage depletions, whereas a much larger set of targets are affected only at full dosage depletions. The presence of a threshold, where the response curves change abruptly, leads to a large change in correlation magnitude across the dosage threshold. In simulations, we recapitulated the three correlation patterns described above with different sets of response curves (Fig. 5c and Methods).

In genetic experiments and genetic association studies, it is common to study the effect of a gene by estimating a single point on its dose–response curve, potentially missing qualitatively different dosage-dependent behavior. An example of this phenomenon is recessivity. More generally, even haploinsufficient genes (such as *SOX9*) can have qualitatively different effects as a function of dosage. These analyses highlight the value of studying allelic series in genetic association studies, and of designing knockdown experiments at clinically relevant dosages.

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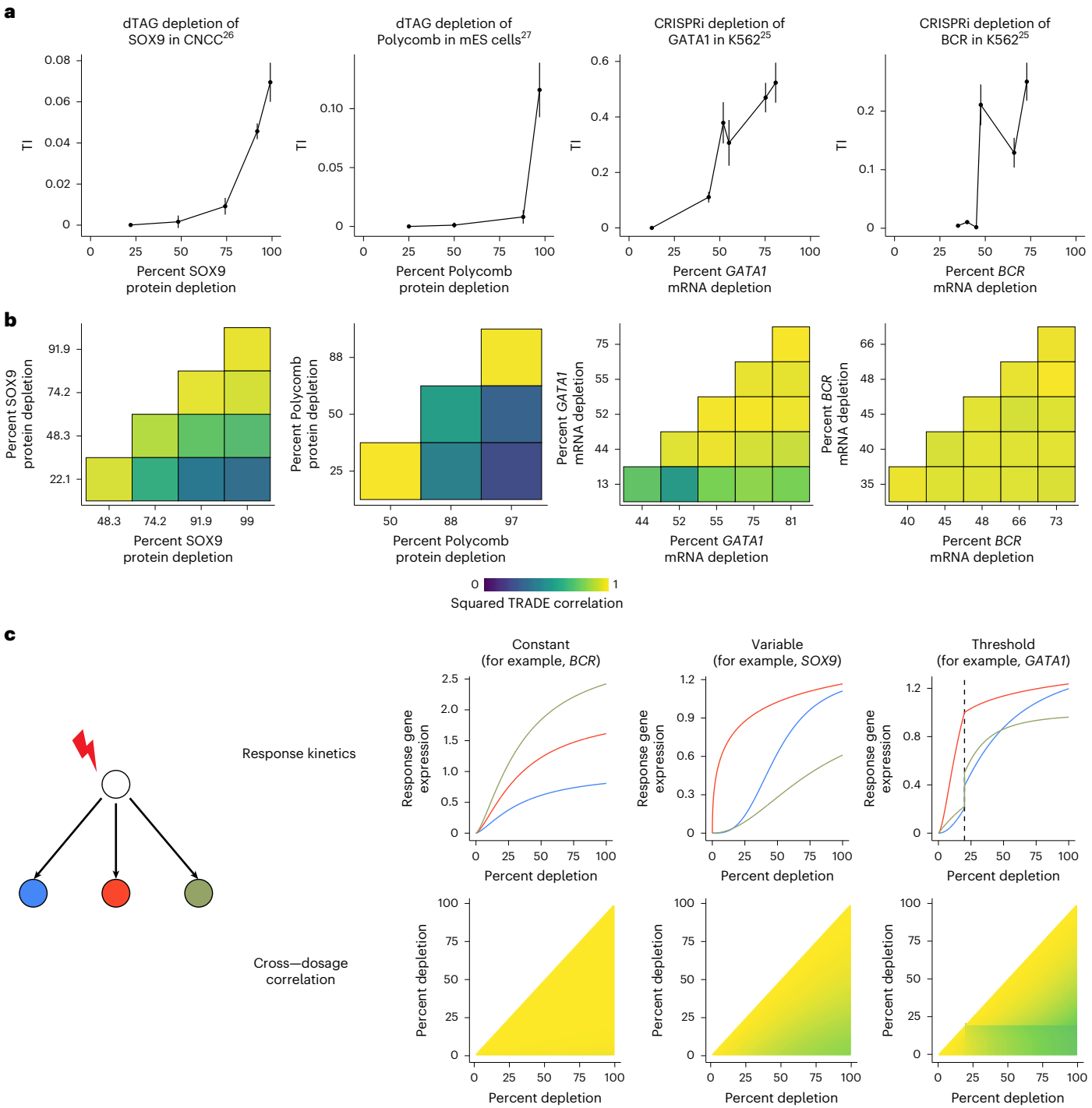


Fig. 5 | Dose–response relationships. a, Relationship between gene dosage and TI across four experiments: dTAG depletion of SOX9 in CNCCs[26]; dTAG depletion of Polycomb in mouse ES cells[27], CRISPRi depletion of GATA1 in K562 cells[25] and CRISPRi depletion of BCR in K562 cells[25]. (*N* per dosage: first panel, seven samples per dosage except for 95% with six samples; second panel, four samples per dosage; third panel, 108, 141, 79, 56, 143 and 52 cells per estimate in ascending order of mRNA depletion; fourth panel, 139, 256, 102, 130, 96 and 151 cells per estimate in ascending order of mRNA depletion.) Error bars denote jackknife standard errors. **b**, Correlations between differential expression effects at different dosages for each experiment. **c**, Observations from a toy model of perturbation effects, demonstrating relationship between response kinetics consistency and resulting pattern of cross-dosage correlations.

Transcriptomic correspondence across neuropsychiatric conditions

Gandal et al.[29] conducted a large-scale differential expression analysis of postmortem brain tissue from people with neuropsychiatric conditions, comparing them with neurotypical controls. They found that differential expression effects were correlated between conditions, and that these correlations paralleled those between the genetic effects on those conditions (the genetic correlation). Because genetic effects are usually causal, this parallel was interpreted as evidence that transcriptomic overlap reflects upstream, disease-causing processes rather than confounding or downstream effects. A notable difference between the transcriptomic and genetic analyses in this study is that the genetic correlation was estimated with a restricted maximum likelihood approach that accounts for sampling variation[30], whereas the transcriptomic

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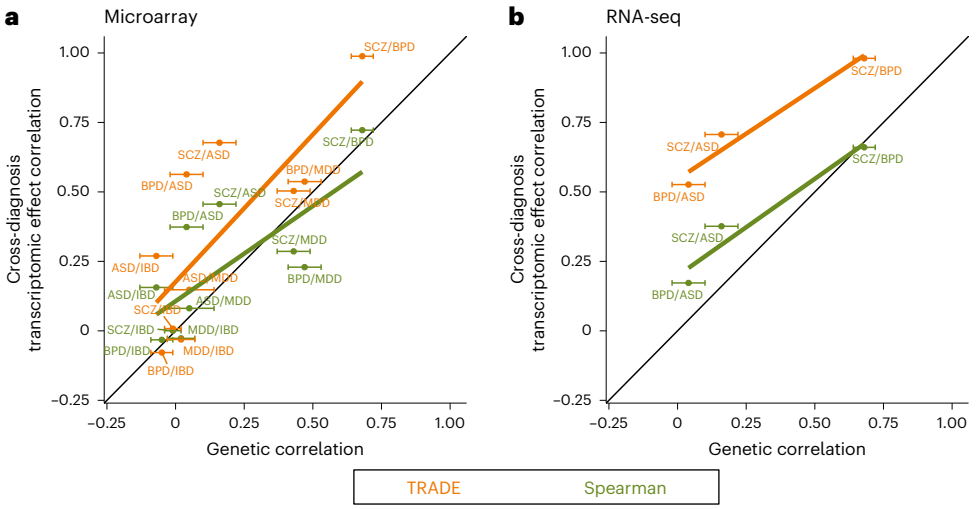


Fig. 6 | Transcriptomic correspondence of neuropsychiatric conditions. **a**,**b**, Estimated TI correlation (orange) compared with Spearman correlations of point estimates (green) across several case/control datasets for neuropsychiatric diagnoses. Estimates for microarray datasets from PsychENCODE (**a**). Estimates for RNA-seq datasets from PsychENCODE (**b**). Error bars denote standard errors of parameter estimates. For details on estimation of genetic correlations and their standard errors, see ref. 49. ASD, autism; BPD, bipolar disorder; IBD, irritable bowel disease; MDD, major depressive disorder; SCZ, schizophrenia.

effect correlation was estimated as the sample Spearman correlation between differential expression point estimates, which is downwardly biased in a sample-size-dependent manner (Supplementary Note 4).

We reanalyzed differential expression summary statistics from this study and estimated the TI correlation between several diagnoses (Fig. 6a,b and Supplementary Table 17). Integrating data from all diagnosis pairs and technologies, we found that TI correlations were substantially larger than sample Spearman correlations, with an increase for nine out of nine psychiatric trait pairs. As a result, unlike the Spearman correlation estimates, the TRADE correlation estimates were larger than the between-condition genetic correlations (Fig. 6a,b). Visualizing the inferred bivariate effect size distributions revealed that, in general, effects are either null in both conditions or positively correlated; there was little evidence of a negatively correlated component or large numbers of condition-specific genes (Supplementary Fig. 24). By contrast, TRADE appropriately estimated lower TI correlations between psychiatric diagnoses and irritable bowel disease, a nonpsychiatric control trait (Supplementary Table 17). One explanation for this difference is that transcriptomic effects are often downstream of condition liability, and these downstream effects are often shared between neuropsychiatric conditions. Another possibility is that there exist confounding factors associated with gene expression and neuropsychiatric diagnoses in general.

One axis of technical variation is that some samples were assayed using RNA-seq, others using microarrays. For three conditions with independent microarray and RNA-seq cohorts in PsychENCODE (autism, bipolar disorder and schizophrenia), we used TRADE to estimate the correlation of transcriptomic effects between assays (Supplementary Table 18). The TI correlation was 0.96, 0.91 and 0.78 for autism, bipolar disorder and schizophrenia, respectively (Supplementary Fig. 25). These estimates indicate that, at least in this study, most differential expression effects replicate between assays.

Discussion

Transcriptomics and differential expression are cornerstones of modern biology. However, a conventional significance-testing framework may conflate biological differences with differences in experimental design or sample size. We showed that these limitations can be addressed by modeling the distribution of differential expression effects explicitly using TRADE. We found that significant genes capture only a fraction of TI in large-scale Perturb-seq experiments. Across cell types, or even replicate experiments, the concordance between estimated effect sizes is attenuated due to sampling variation but, in many cases, the true effect sizes are highly concordant. In dose–response experiments, we found that dosage affects not only the magnitude of the transcriptome-wide effect, but also the genes that are affected. In a case–control as opposed to a perturbational dataset, we found that the same advantages apply, and that our approach changes the interpretation of a key analysis of neuropsychiatric conditions.

The ubiquity of small differential expression effects is connected to an existing dichotomy in the field, between approaches that test for differential expression of single genes (for example, DESeq2)[12] and those that test for differential abundance of cellular states (for example, covarying neighborhood analysis (CNA)[31]). These methods approach differential expression with distinct priors: that changes in expression will be restricted largely to a small number of genes with large effect, or that changes in expression will be spread across many hundreds or thousands of genes, reflecting a change in cell state. Estimates from TRADE, in particular π_{DEG}, can contextualize these approaches by quantifying the degree to which differential expression is concentrated in specific target genes versus spread across the transcriptome.

In addition to studying perturbations, an important application of differential expression analysis is to understand differences between cell types. Many analyses of cell-type variation require a distance metric—a scalar summary of the transcriptomic difference between groups of cells—and many such metrics have been proposed[32]. TI may be a suitable distance metric for such analyses, as it is unbiased at finite sample size (unlike the commonly used Euclidean distance; Supplementary Note 6) and can be computed from differential expression summary statistics. Indeed, we found that, compared with Euclidean distance, TI produced a more coherent cell-type hierarchy of peripheral blood mononuclear cells in the OneK1K dataset[33,34] (Supplementary Fig. 26, Supplementary Table 19 and Methods). However, a limitation of TRADE is that it relies upon predefined labels and cannot be used to cluster cells into cell types.

For genetic perturbations, parameters such as TI are probably driven by the pattern of causal regulatory connections between genes, that is, the gene regulatory network (GRN). Inference of GRNs from single-cell measurements is a challenging and unsolved technical problem[35]. We speculate that, just as inferring TI is easier than inferring gene-specific effect sizes, estimating global features of the GRN may be easier than identifying individual edges. This could be

achieved by pairing TRADE with a model relating the distribution of differential expression effects to GRN features such as the degree distribution or modularity. We speculate that the true GRN is densely interconnected with relatively low modularity, based on our observation that virtually all high TI perturbations also affect a large number of genes, approaching the number of genes that are expressed (Supplementary Note 3).

A key limitation of TRADE is that it currently uses only a simple readout from single-cell RNA-seq experiments, the pseudobulk mean RNA expression level. On the one hand, modeling pseudobulk counts has been demonstrated to prevent pseudoreplication bias[36,37]. On the other hand, the transcriptional state of individual cells may vary in ways that are poorly captured in pseudobulk (for example, due to the presence of several cell types)[38]. In addition, some biological processes are better assayed using alternative modalities, including mRNA splicing, chromatin state, protein level and imaging, all of which are now being studied at scale with single-cell CRISPR screens[39–44]. Expanding TRADE to other phenotypic readouts may be straightforward but will require careful consideration of how to model measurement noise and whether the Gaussian model assumed by ash is appropriate.

An emerging goal of functional genomics is the generation of perturbational cell atlases across several cellular contexts[45,46]. However, as with all screening methods, there is a tradeoff where the number of assayed perturbations is ultimately constrained by experimental cost. TRADE shifts the balance in this tradeoff by allowing stable quantification of highly informative metrics including TI, correlations between perturbations and context-dependent effects at much shallower sampling depths. Combined with developments in screen compression[47] and cheaper sequencing technologies[48], our method suggests a productive path toward massive scale perturbational atlases.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41588-025-02169-3.

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Methods

Ethical statement

This research falls under the category of exempt, nonhuman-participant research.

Experimental model and subject details: Perturb-seq of essential genes in Jurkat and HepG2

Cell line generation and maintenance. All cell lines were grown at 37 °C in the presence of 5% CO_2 in standard tissue culture incubators.

A CRISPRi Jurkat cell line expressing dCas9-BFP-KRAB (KOX1-derived) was obtained from the UC Berkeley Cell Culture Facility (clGI1) and was used for growth screens. A second CRISPRi Jurkat cell line expressing the optimized UCOE-EF1α-Zim3-dCas9-P2A-mCherry CRISPRi construct was generated as previously described[50] and was used for Perturb-seq. Jurkat cells were grown in RPMI-1640 medium with 25 mM HEPES, 2.0 g l^{-1} $NaHCO_3$ and 0.3 g l^{-1} L-glutamine (Gibco) supplemented with 10% (v/v) standard FBS, 2 mM glutamine, 100 units ml^{-1} penicillin and 100 µg ml^{-1} streptomycin (Gibco).

A CRISPRi HepG2 cell line expressing UCOE-EF1α-dCas9-BFP-KRAB (KOX1-derived) was obtained from Torres et al.[51], and was used for both growth screens and Perturb-seq. HepG2 cells were grown in EMEM with 1.5 g l^{-1} $NaHCO_3$, 110 mg l^{-1} sodium pyruvate, 292 mg l^{-1} L-glutamine (Corning) supplemented with 10% (v/v) standard FBS, 100 units ml^{-1} penicillin and 100 µg ml^{-1} streptomycin (Gibco).

HEK293T cells were used for generation of lentivirus. HEK293T cells were grown in DMEM with 25 mM D-glucose, 3.7 g l^{-1} NaHCO3, 4 mM L-glutamine (Gibco) supplemented with 10% (v/v) standard FBS, 2 mM glutamine, 100 units ml^{-1} penicillin and 100 µg ml^{-1} streptomycin (Gibco).

Lentiviral production. To produce lentivirus, HEK293T cells were cotransfected with transfer plasmids and standard packaging vectors expressing VSV-G, Gag/Pol, Rev and Tat using TransIT-LTI Transfection Reagent (Mirus). Viral supernatant was harvested 2 days after transfection and frozen at −80 °C before transduction.

Library design and growth screens. Dual-single guide RNA (sgRNA) CRISPRi lentiviral libraries were described previously[6]. In brief, a preliminary sgRNA library (dJR058, n = 2,291 dual-sgRNA elements) with even representation of all dual-sgRNA constructs was used for growth screens. This library contains a single dual-sgRNA construct targeting (1) 20Q1 Cancer Dependency Map common essential genes (https://depmap.org/portal/download/) and (2) 5% nontargeting control sgRNAs cloned into pJR101 (Addgene, cat. no. 187241). A second sgRNA library (dJR092, n = 2,688 dual-sgRNA elements; Supplementary Table 20) that adjusted the representation of sgRNAs to decrease dropout of essential genes was used for Perturb-seq experiments. The sgRNA abundance was corrected according to the effects observed in growth screens (described below); for example, a guide with roughly fourfold depletion in growth screens was fourfold overrepresented in dJR092. This library also included sgRNAs targeting a number of additional genes with interesting phenotypes in the K562 genome-wide Perturb-seq dataset.

Pooled growth screens in Jurkat cells were performed by transducing Jurkat cells expressing dCas9-BFP-KRAB (clGI1) with the dual-sgRNA CRISPRi lentiviral library, dJR058. Screens were performed in biological replicate maintaining a coverage of >1,000 cells per library element for the duration of the screen. Cells were transduced by spinfection (1,000g) with polybrene (8 µg ml^{-1}; Sigma-Aldrich) to obtain an infection rate of 10–20%. On day 3 after transduction, cells were sorted to near purity by FACS (FACSAria2; BD Biosciences) using green fluorescent protein (GFP) as a marker for sgRNA vector transduction. On day 7 after transduction, an aliquot of cells was harvested for sequencing to compare sgRNA abundances to the plasmid library.

Pooled growth screens in HepG2 cells were performed by transducing HepG2 cells expressing dCas9-BFP-KRAB with the dual-sgRNA CRISPRi lentiviral library, dJR058. Screens were performed in biological replicate maintaining a coverage of >1,000 cells per library element for the duration of the screen. Cells were transduced by plating in viral supernatant with polybrene (8 µg ml^{-1}; Sigma-Aldrich) to obtain an infection rate of 30–40% based on GFP measurement by FACS (FACSAria2, BD Biosciences). On day 7 after transduction, an aliquot of cells was harvested for sequencing to compare sgRNA abundances with the plasmid library. Library preparation and sequencing of growth screens followed the protocol previously described[6].

Perturb-seq experiments, library preparation and sequencing. For the Jurkat Perturb-seq experiment, Jurkat cells expressing Zim3-dCas9-P2A-mCherry were transduced with dJR092 library lentivirus by spinfection (1,000g) with polybrene (8 µg ml^{-1}; Sigma-Aldrich) with a targeted low infection rate of ~10%. This low rate was chosen to reduce the chances of a single cell being infected by several viruses. Cells were maintained at a coverage of >1,000 cells per library element for the duration of the screen. On day 3 after transduction, an infection rate of 7% was measured based on GFP as a marker of transduction, and cells were sorted to near purity by FACS (FACSAria2, BD Biosciences). On day 7 after transduction, cells were measured to be >90% GFP positive (Attune NxT, ThermoFisher) and were prepared for single-cell RNA-seq by resuspension in 1× PBS with 0.04% BSA as detailed in the 10x Genomics Single Cell Protocols Cell Preparation Guide (10x Genomics, cat. no. CG00053 Rev C). Cells were separated into droplet emulsions using the Chromium Controller (10x Genomics) with Chromium Single-Cell 3′ Gel Beads v.3 (10x Genomics, cat nos. PN-1000075 and PN-1000153) across 56 GEM groups following the 10x Genomics Chromium Single Cell 3′ Reagent Kits v.3 User Guide with Feature Barcode technology for CRISPR Screening (cat. no. CG000184 Rev C) with the goal of recovering ~15,000 cells per GEM group before filtering.

For the HepG2 Perturb-seq experiment, HepG2 cells expressing dCas9-BFP-KRAB were transduced with dJR092 library lentiviral supernatant with polybrene (8 µg ml^{-1}; Sigma-Aldrich) with a targeted low infection rate of ~10%. Cells were maintained at a coverage of >1,000 cells per library element for the duration of the screen. On day 3 after transduction, an infection rate of 7% was measured based on GFP as a marker of transduction and cells were sorted to near purity by FACS (FACSAria2, BD Biosciences). On day 7 after transduction, cells were dissociated using Accutase (StemCell Technologies) for 30 min and resuspended in 5 mM EDTA-PBS. To decrease cell doublets in the single-cell RNA-seq, GFP-positive singlets were isolated by FACS (FACSAria2, BD Biosciences) with a final cell population measured to be ~90% GFP positive. Cells were then prepared for single-cell RNA-seq by resuspension in 1× PBS with 0.04% BSA as detailed in the 10x Genomics Single Cell Protocols Cell Preparation Guide (10x Genomics, cat. no. CG00053 Rev C). Cells were separated into droplet emulsions using the Chromium Controller (10x Genomics) with Chromium Single-Cell 3′ Gel Beads v.3 (10x Genomics, cat. nos. PN-1000075 and PN-1000153) across 56 GEM groups following the 10x Genomics Chromium Single Cell 3′ Reagent Kits v.3 User Guide with Feature Barcode technology for CRISPR Screening (cat. no. CG000184 Rev C) with the goal of recovering ~15,000 cells per GEM group before filtering.

For library preparation, samples were processed according to 10x Genomics Chromium Single Cell 3′ Reagent Kits v.3 User Guide with Feature Barcode technology for CRISPR Screening (cat. no. CG000184 Rev C) with magnetic selections conducted on an Alpaqua Catalyst 96 plate (cat. no. A000550). For sequencing, mRNA and sgRNA libraries were pooled to avoid index collisions and sequenced on a NovaSeq 6000 (Illumina) according to the 10x Genomics User Guide.

Quantification and statistical analysis

Perturb-seq in Jurkat and HepG2: alignment, cell calling, sgRNA assignment and cell filtering. As described previously[6], Cell Ranger v.4.0.0 software (10x Genomics) was used for scRNA-seq and sgRNA

alignment, collapsing reads to UMI counts, and cell calling. The 10x Genomics GRCh38 version 2020-A genome build was used as a reference transcriptome. For sgRNA assignment, reads were first downsampled by GEM group to produce a more even distribution of the number of reads per cell, with an upper threshold of 3,000 mean mapped reads per cell in the Jurkat experiment and 2500 mean mapped reads per cell in the HepG2 experiment. Guide calling was performed with a Poisson-Gaussian mixture model as previously described[52], with only cells bearing two sgRNAs targeting the same gene or a single sgRNA used for downstream analysis. Cells were filtered for quality to remove cells with low unique molecular identifier (UMI) content (Jurkat, >1,750 UMIs; HepG2, >3,000 UMIs) and high mitochondrial RNA content (Jurkat, <14%; HepG2, <18%). These filters removed 7,408 cells from the Jurkat-Essential experiment (262,956 cells retained) and 15,952 cells from the HepG2-Essential experiment (145,473 cells retained). A similar procedure was performed as previously described for the K562-GenomeWide, K562-Essential and RPE1-Essential datasets[6].

Modeling transcriptome-wide responses. In a differential expression experiment, read depth is quantified for each gene and each cell or sample. The resulting counts are typically modeled as following a distribution (for example, negative binomial) whose mean may differ between two conditions, and the difference is quantified as logFC, defined as the difference between the logarithm of population expression means between the two conditions. In a typical experiment, the logFC can be interpreted as the effect size of a condition or perturbation on a gene, and many biological questions are related to the distribution of true effect sizes across genes. However, we only observe the distribution of estimated effect sizes. These estimates can be modeled as the sum of two distributions: the distribution of true effect sizes and the sampling distribution in the experiment. TRADE is a method to disentangle these components.

To do so, TRADE uses ash[13] to estimate the effect size distribution from differential expression summary statistics, which we compute with DESeq2 (ref. 12). DESeq2 fits a regularized negative binomial generalized linear model, sharing information across genes to improve overdispersion estimates. It has been applied to many different contrasts, including genetic perturbations and different cell types. ash models an effect size distribution by learning the weights of a flexible mixture model using maximum likelihood. ash incorporates s.e. estimates into the inference procedure, effectively downweighting noisier logFC estimates, such as those for very low-expressed genes.

We analyzed the estimated previous distribution, rather than using it to perform shrinkage on each individual effect, which is the original goal of the ash method. Our estimates could be used in this way, although we caution that slight downward bias in our TI estimates (Supplementary Fig. 1) may lead to slight overshrinkage for underpowered perturbations. We briefly explored TRADE-generated posterior-mean effect size estimates in simulations, finding that these posterior-mean estimates had lower mean squared error than DESeq2 effect size estimates (Supplementary Fig. 5), suggesting that this is a promising direction for future research.

Notably, because the datasets presently analyzed do not contain within-experiment cell-type heterogeneity, TRADE at present does not model cell-type variability. For such datasets, several approaches could be taken, including (1) stratifying differential expression analysis by cell type and conducting separate TRADE analyses for each cell type, (2) including discrete cell-type annotations as fixed effects in the differential expression model, or (3) including continuous axes of transcriptomic variability (that is principal components) as fixed effects in the differential expression model. We suspect that the appropriate method will depend on the particular scientific question.

Features of the effect size distribution. The TI is defined as the variance of the distribution of differential expression effect sizes, in units of $\log_2$FC. The TI captures the overall degree of transcriptomic change across a contrast of interest, for example a perturbation. TI is in interpretable units of $(\log_2\text{FC})^2$. For example, if a hypothetical perturbation affects all genes with normally distributed effect sizes, and the TI is 0.25, it means that a typical gene has an effect size of 0.5.

Beyond this simplistic model, a large TI may arise either because a perturbation has a large effect on a few genes, or because it has smaller effects on many of them. To distinguish between these possibilities, we define the effective number of DEGs (π_{DEG}) as:

$$\pi_{\text{DEG}} = \frac{3M}{\kappa} \quad \kappa = \frac{E[\beta^4]}{E[\beta^2]^2}$$

where M is the number of genes with measured expression and κ is the kurtosis (normalized fourth moment) of the inferred effect size distribution. If a perturbation has a large effect on only a few genes, then κ is large, and π_{DEG} is small. Conversely, if a perturbation affects all genes with normally distributed effect sizes, then κ equals 3 and π_{DEG} is equal to the number of genes (Supplementary Fig. 11).

Different sets of genes may be enriched or depleted for differential expression. We define the perturbation response enrichment of a geneset as:

$$\text{Enrichment}_{\text{response}}(\text{geneset}) = \frac{\text{Var}(\beta_{\text{gene}}|\text{gene} \in \text{geneset})}{\text{Var}(\beta_{\text{gene}})}.$$

We estimate the numerator by applying ash to genes in the geneset, and we estimate the denominator by applying it to all genes. This approach is expected to be approximately unbiased for most genesets. However, we do also use it to estimate the fraction of signal in FDR-significant genes; we note that such estimates are expected to be upwardly biased by winner's curse. In Fig. 3, we report the mean perturbation response enrichment across perturbations.

In addition to the perturbation response enrichment, we also estimate the perturbation impact enrichment. If TI_i is the TI of perturbation i, the perturbation impact enrichment is:

$$\text{Enrichment}_{\text{impact}}(\text{geneset}) = \frac{n_{\text{perturbations}} \times \sum_{\text{geneset}} \text{TI}_i}{n_{\text{geneset}} \times \sum_{\text{all perturbations}} \text{TI}_i}.$$

We estimate this quantity by substituting the estimated TI for the true TI.

For all TRADE estimates, we estimate P values using a Wald test.

TRADE implementation details (univariate). In brief, to model the effect size distribution, we used ash to fit the following mixture model:

$$z_g \sim \text{Multinomial}([k]; \pi_1, \ldots, \pi_k)$$
$$\beta_g | z_g = i \sim \text{Unif}(0, b_i)$$
$$\widehat{\beta_g} \sim N(\beta_g, \hat{s}_g^2)$$

Where $\hat{\beta}_g$ is the estimated $\log_2$FC, β_g is the true $\log_2$FC, $\hat{s}_g$ is the estimated s.e., z_g matches gene g to a mixture component uniform distribution i with one extremum at 0 and the other at b_i and π_i are the weights for components of the mixture distribution.

ash fits this model with interior point optimization methods (for details, see ref. 13). For mixture components, we used a fine grid of uniform distributions with one extremum at zero as mixture components ('half-uniform') rather than zero-centered uniform mixture components, to allow for estimation of asymmetric effect size distributions; Stephens[13] found that these this model is sufficiently flexible to model realistic effect size distributions.

We largely used ash with default settings. We made two modifications: (1) we restricted the range of mixture components to the smallest and largest observed effect size, rather than the default behavior of

c(−Inf, Inf), to improve computational efficiency. (2) We used a uniform rather than null-biased prior; using a null-biased prior is crucial for accurately computing the local false sign rate, but is less important for estimating the effect size distribution itself. We removed the null-biased prior to prevent bias in our distribution estimate.

Simulations. For detailed description of simulations, see Supplementary Note 1.

Genome-wide Perturb-seq. We analyzed data from five large-scale Perturb-seq experiments, including three from Replogle et al.[6] (K562-GenomeWide, K562-Essential, RPE1-Essential) and two that are new (Jurkat-Essential, HepG2-Essential) ('Data availability'). We generated differential expression summary statistics (that is, $\log_2$FC and s.e.) for each perturbation as follows. We generated a per-batch ('gem-group') pseudobulk dataset, summing counts across control cells (that is, cells carrying a nontargeting guide RNA) and perturbed cells (that is, cells carrying a guide RNA against a particular gene) in each batch. We estimated differential expression effects from this pseudobulk dataset using DESeq2, with an identical model as in our simulations (see above). We computed *P* values with the likelihood ratio test as implemented in DESeq2.

We modeled each batch as a fixed effect, and DESeq2 scales poorly with the number of covariates. This presented serious challenges only for the K562-GenomeWide dataset, which had 272 batches. To circumvent this issue, we analyzed the K562-GenomeWide dataset in four 'mega-batches' of 68 batches each (assigned to mega-batches based on their numerical indices). We then produced final differential expression summary statistics for the whole experiment by meta-analyzing the resulting four sets of $\log_2$FC estimates using inverse variance weighted meta-analysis.

The number of pseudobulk units in our differential expression is twice the number of batches (that is, one pseudobulk observation for control and perturbed, for each batch). In the K562-Essential experiment, there were 48 batches. In the RPE1-Essential experiment, there were 56 batches. In the Jurkat-Essential experiment, there were 55 batches. In the HepG2-Essential experiment, there were 56 batches. In the K562-GenomeWide experiment, there were 272 batches.

In single-cell differential expression analyses, a failure to account for biological variability across supracellular sampling units (that is animal, donor, and so on) can lead to overestimation of effects; this phenomenon is referred to as 'pseudoreplication bias'[36,37]. In our analyses, the pseudobulk observation units reflect different sequencing batches drawn from the same cell line. Because there is no variability due to supracellular sampling unit (that is animal, donor, and so on), the influence of pseudoreplication bias is expected to be minimal.

Gene annotations. For our enrichment analyses in the K562-GenomeWide dataset, we used the following gene annotations ('Data availability'): expression level, from the K562-GenomeWide dataset itself; growth effect, downloaded from the K562 CRISPR growth screen in the Cancer DepMap project; loss of function observed over expected upper fraction, from the gnomad v.2 resource[17]; nuclear and cytoskeletal localization, from the COMPARTMENTS database[18]; DE prior,from the supplementary information of ref. 19; stably expressed genes, from ref. 20; on-target knockdown: from DESeq2.

For the quantitative annotations, we generated two annotations—top 10% and bottom 50%—for enrichment analyses.

Transcriptome-wide analysis of differential expression (bivariate). Given two sets of differential expression summary statistics (for example, $\log_2$FC and s.e. computed with DESeq2), we estimated the joint distribution of effect sizes using 'mash'[53]. mash fits a mixture of multivariate normal distributions to model the joint distribution of effect sizes across an arbitrary number of experiments, for example, expression quantitative trait loci from tens of tissues; we used mash to model bivariate effect size distributions, with a particular interest in estimating the correlation of effects between two perturbations.

In brief, mash finds the weights π that maximize the following likelihood:

$$z_g \sim \text{Multinomial}([k]; \pi_1, \ldots \pi_k)$$
$$\beta_g | z_g = i \sim \text{MVN}(0, U_i)$$
$$\hat{\beta}_g \sim \text{MVN}\left(\beta_g, S_g^2\right)$$

where $\hat{\beta}_g$ is a two-element vector of estimated effect sizes, β_g is a two-element vector of true effect sizes, S_g^2 is the sampling covariance matrix of the true effect sizes, z_g matches gene g to a mixture component multivariate normal distribution parameterized by fixed covariance matrix U_i and π_i is the weight for the component i of the mixture distribution.

We choose S_g to be a diagonal matrix with diagonal entries equal to the variance of the individual estimated effects. This choice is appropriate when each estimate is derived from a different experiment, which is the case in our analyses.

Selecting the covariance matrices U_i is a crucial step in this analysis. By default, mash recommends a combination of 'canonical' (that is, reflecting simple correlation patterns) and data-derived (that is, from factorization of the observed data matrix) covariance matrices, across a range of scaling factors. We used these mash default covariance matrices, and added several more matrices comprising an 'adaptive grid.' We did so because, whereas mash was designed primarily for multivariate experiments with several conditions where specifying all possible covariance patterns is not feasible, we are interested in the bivariate case, where doing so is feasible.

We obtain this adaptive grid of covariance matrices by first running univariate ash in each condition, with half-normal mixture components. We then retain the component variances with nonzero weight for each distribution. Then, for each combination of variances, we create covariances matrices with several covariance values corresponding to a grid of correlations between −1 and 1 (in our experiments, 21 correlation values was a sufficiently dense grid). This procedure produces a set of covariance matrices that attempt to tile all possible bivariate relationships between the two perturbations.

Identification of genes with cell-type-specific perturbation effects. To identify perturbations with exceptionally large TI in one cell type, we regressed log-transformed TI estimates from each cell type on the median log-transformed TI across all four cell types. This regression included 2,050 perturbations, excluding three common essential genes that had zero TI in at least one cell type. We then defined perturbations with cell-type specific effects as perturbations with a standardized residual from this regression greater than 1.64, that is, corresponding to a *P* value of 0.1.

Notably, this regression included fitted parameters for both intercept and slope, meaning that cell-type-specific effects were not identified only because one cell type exhibits stronger effects overall.

Clustering genetic perturbations across cell types. Visualization of the relationship between TI correlation within and between each pair of more-similar cell types (Fig. 4d) motivated us to cluster perturbations based on these values with a bivariate Gaussian mixture model. We fit a bivariate Gaussian mixture model using an expectation-maximization algorithm as implemented in the mclust package in R. The resulting mixture components reflected the visually apparent clusters from Fig. 4d, that is, including one component with relatively high correlations within and outside of similar cell types (mean correlation within, 0.75; mean correlation outside, 0.66) and one component with lower

correlations outside of similar cell types (mean correlation within, 0.61; mean correlation outside, 0.35). We assigned each genetic perturbation to one of the two components based on the posterior probabilities from this model.

Perturb-seq with attenuated guide RNAs. We downloaded publicly available processed scRNA-seq data from ref. 25. Full details are available in the primary manuscript describing this dataset. This data structure is largely identical to those described above from Replogle et al.[6], with several guides (with several targeting each of 25 essential genes) arrayed across three batches.

To analyze this dataset, we used an approach identical to that of the genome-wide and essential-wide experiments of Replogle et al.[6], performing a batch pseudobulk analysis with DESeq2. To harmonize this analysis with that of Replogle et al.[6], we limited the measured genes analyzed to those with an average expression level of 0.01 UMIs across cells. To estimate standard errors, we used a block-jackknife across cells with 100 blocks.

We estimated the degree of ontarget knockdown using the $\log_2$FC for the target gene from DESeq2.

dTAG depletion of SOX9. Gene-wise RNA counts were downloaded from the Zenodo archive accompanying ref. 26, and differential expression analysis was conducted using the script from the same repository. In brief, RNA was sequenced from bulk samples of human embryonic-stem-cell-derived human neural crest cells with varying concentrations of dTAG targeting SOX9. RNA-seq data were aligned with Salmon[54], and differential expression analysis was carried out with DESeq2 with differentiation batch as a covariate; s.e. for the TRADE analysis was computed using a sample jackknife.

dTAG depletion of polycomb repressive complex. Gene-wise RNA counts were downloaded from the GEO repository accompanying ref. 27. In brief, RNA was sequenced from bulk samples of mouse embryonic stem cells with varying concentrations of dTag targeting RING1b and EED. RNA-seq data were aligned with kallisto[55], and differential expression analysis was carried out with DESeq2; s.e. for the TRADE analysis was computed using a sample jackknife.

Simulations of dose–response curves. To simulate correlation of perturbation effects across dosage levels, we simulated 10,000 measured gene expression profiles with varying perturbation effects across genes. The response function of each gene was simulated with a Hill equation:

$$Y = Y_{\max}\mathrm{FC}_0 + Y_{\max}(1-\mathrm{FC}_0)\frac{1}{1+\left(\frac{a}{x}\right)^b}$$

where Y is the expression level of the target gene at dosage level x of the perturbed gene, $Y_{\max}$ is the expression level of the target gene at full dosage of the perturbed gene, FC_0 is the FC of the target gene associated with full depletion of the perturbed gene, a is the concentration associated with half-maximal response and b is the 'Hill coefficient' or the degree of cooperativity.

For all simulations, across genes, $Y_{\max}$ was drawn from the normal distribution $N(100,5)$ and FC_0 was drawn from the normal distribution $N(0,2)$ (that is, infinitesimal architecture).

For the 'Constant' simulation, a and b were constant (50 and 0.5, respectively). For the 'Variable' simulation, a was drawn from the uniform distribution Unif(10.90) and b was drawn from the uniform distribution Unif(0.1,5). For the threshold simulation, a and b were drawn from these uniform distributions two times independently, to compute curves before and after a threshold of 20.

From the simulated response kinetic curves, $\log_2$FC values were computed, and correlated across dosage levels.

Case–control differential expression in neuropsychiatric disorders. We downloaded case–control, RNA-seq and microarray-based differential expression summary statistics for the PsychENCODE dataset from ref. 29 for autism, schizophrenia, bipolar disorder, major depressive disorder and irritable bowel disease. Following ref. 29, we used the estimates of cross-disorder genetic correlation from the Psychiatric Genomics Consortium[56].

Estimating cell-type hierarchies in the OneK1K dataset. We downloaded count-based sequencing data from the OneK1K cohort[33], using the postpublication quality control of this dataset by Rumker et al.[57]. We excluded one person who had cells present in several batches. We generated a pseudobulk dataset by summing the counts of each person, for each of the 28 peripheral blood mononuclear cell types (that is, excluding erythrocytes, hematopoietic stem and progenitor cells, and platelets) identified by Rumker et al.[57]. We generated a pseudobulk dataset by summing counts within each person for each cell type. We then used DESeq2 to estimate differential expression between each pair of cell types, with batch as a covariate. We then used TRADE to estimate the TI between pairs of cell types.

For the Euclidean distance analysis, we took an approach similar to the above, but fit a DESeq2 model with only an intercept term to the pseudobulk data from each cell type, to estimate the mean expression of each gene. For each cell type, we then normalized these mean expression profiles by converting to 'counts-per-10,000' units (cp10k), adding 1 and log-transforming. We then computed the Euclidean distance between each pair of normalized mean cell-type expression profiles.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Raw sequencing data are deposited on SRA under BioProject PRJNA1100571. Aligned sequencing data and processed single-cell populations are available on GEO at GSE264667. Perturb-seq data from Replogle et al.[6] can be accessed at https://gwps.wi.mit.edu/ (raw data available at SRA under BioProject PRJNA831566). dTAG data from Naqvi et al.[26] are available at Gene Expression Omnibus (GEO) under accession number GSE205904. dTAG data from Weber et al.[27] are available at GEO under accession number GSE145016. Summary statistics from the PsychENCODE consortium are available at https://github.com/mgandal/Shared-molecular-neuropathology-across-major-psychiatric-disorders-parallels-polygenic-overlap; Raw data are all available at Synapse under accession number syn4921369. RNA-seq data from the OneK1K dataset are available at GEO under accession number GSE196830.

Code availability

The TRADE method, with accompanying documentation, is publicly available as an R package at https://github.com/ajaynadig/TRADEtools. The publication version of this package is available as a persistent repository via Zenodo at https://doi.org/10.5281/zenodo.14993815 (ref. 58).

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Author contributions

A.N., E.B.R. and L.J.O. conceived the study. J.M.R. and A.N.P. collected new data. J.S.W. supervised new data collection. A.N., M.M. and L.J.O. conducted analyses. A.N., J.M.R., M.M. and L.J.O. wrote the manuscript. S.A.M. provided additional project supervision.

Competing interests

J.S.W. declares outside interest in 5 AM Venture, Amgen, Chroma Medicine, KSQ Therapeutics, Maze Therapeutics, Tenaya Therapeutics, Tessera Therapeutics, Ziada Therapeutics and Third Rock Ventures. J.M.R. consults for Third Rock Ventures and Maze Therapeutics, and is a consultant for and equity holder in Waypoint Bio. The remaining authors declare no competing interests.

Additional information

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